

8-Day Budget Calculator Project Plan

✓ Project Scope (Condensed but Complete)

Must-Have Features:

- User input for income and expenses
- Calculates totals and savings
- Saves/loads data with JSON
- Console summary report
- Expense pie chart using `matplotlib`

Optional (if time permits):

- GUI using `tkinter`
 - Export to PDF or TXT
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Day-by-Day Plan

Day 1: Input System + Data Storage Design

- Build input functions: `add_income()`, `add_expense()`
- Use lists of dictionaries for storage
- Create a main CLI menu loop

Day 2: Calculation Functions

- Write functions to calculate:

- Total income
 - Total expenses
 - Savings (income – expenses)
- Print basic output from totals

Day 3: Save & Load with JSON

- Use `json` module to:
 - Save session data to `budget_data.json`
 - Load existing data at startup
- Add basic error handling

Day 4: View/Edit/Delete Entries

- Display income/expenses with indexes
- Implement `edit_entry()` and `delete_entry()` functions

Day 5: Summary Report

- Print a formatted report with:
 - All incomes and expenses
 - Totals and savings summary

Day 6: Matplotlib Visualization

- Generate pie chart of expenses by category
- Optional: Bar chart for income vs expenses
- Save chart as image (e.g., `budget_chart.png`)

Day 7: Input Validation + Cleanup

- Prevent:
 - Blank entries
 - Negative values
 - Non-numeric inputs
- Clean up redundant code
- Prepare a sample data file

Day 8: Testing + README

- Test all features: input, calc, save/load, charts
- Write `README.md` with:
 - Features
 - Setup instructions
 - Dependencies (`json`, `matplotlib`)
- Optional: Start basic `tkinter` GUI if time allows